ALGORITHMS

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Outline

History

Concepts

- Definition
- How to develop an algorithm
- Essential features

Three Constructs

Algorithm Representation

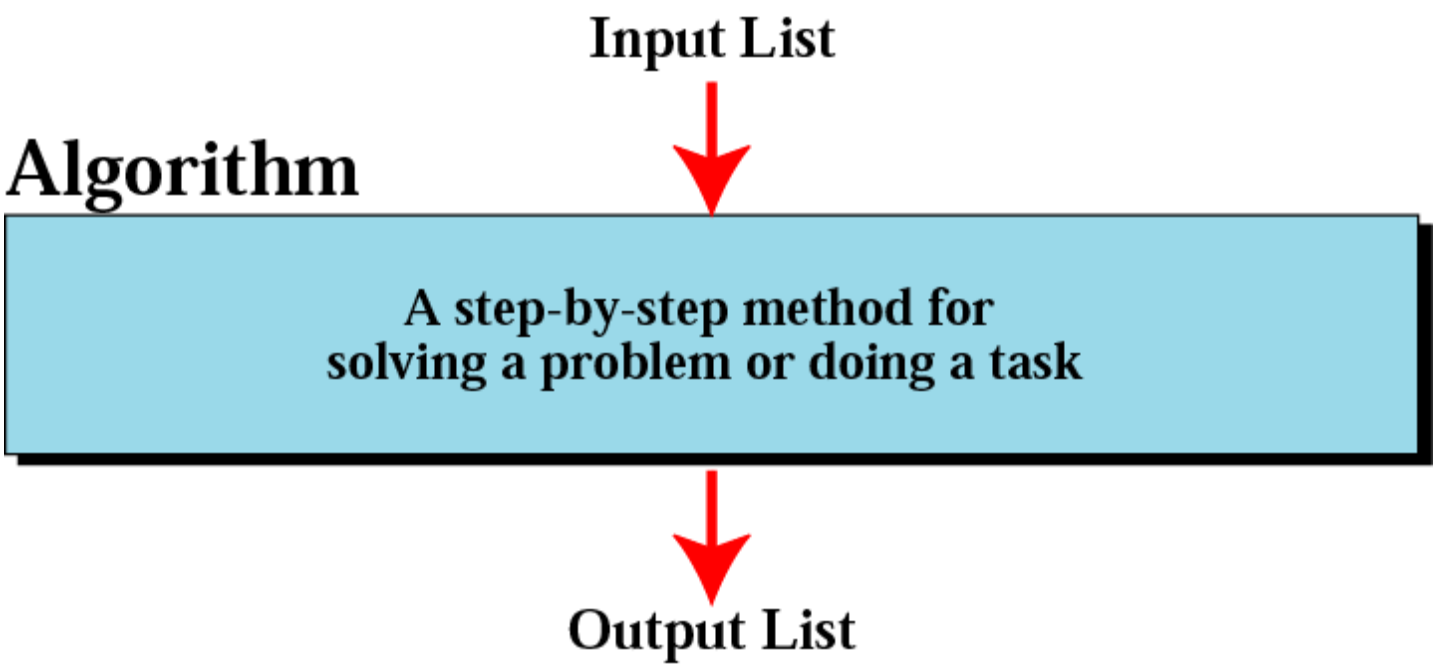
- Pseudocode
- Flowchart

Basic algorithms

History of Algorithm

- It was derived from the name of Muslim mathematician Abu Ja`far Mohammad Ibn Musa al- Khowarizmi
- Khowarizm is a place in Uzbekistan
- al-Khowarizmi was born in the 8th century
- He settled in Baghdad along with his family and did his education there.
- He died in Baghdad in 834 A.D
- One of his book is “al Kitab al mukhatasar fi hisab al-jabr wa`l-muqabila”
- Algebra was derived from “al-jabr” (his book name)
- He played a big role in inventing digits from 0-9

Informal definition



Formal definition

- An algorithm is thus a sequence of computational steps that transform the input into output.

OR

- Step by Step procedure to solve a problem in a specific time.

PROPERTIES OF ALGORITHMS

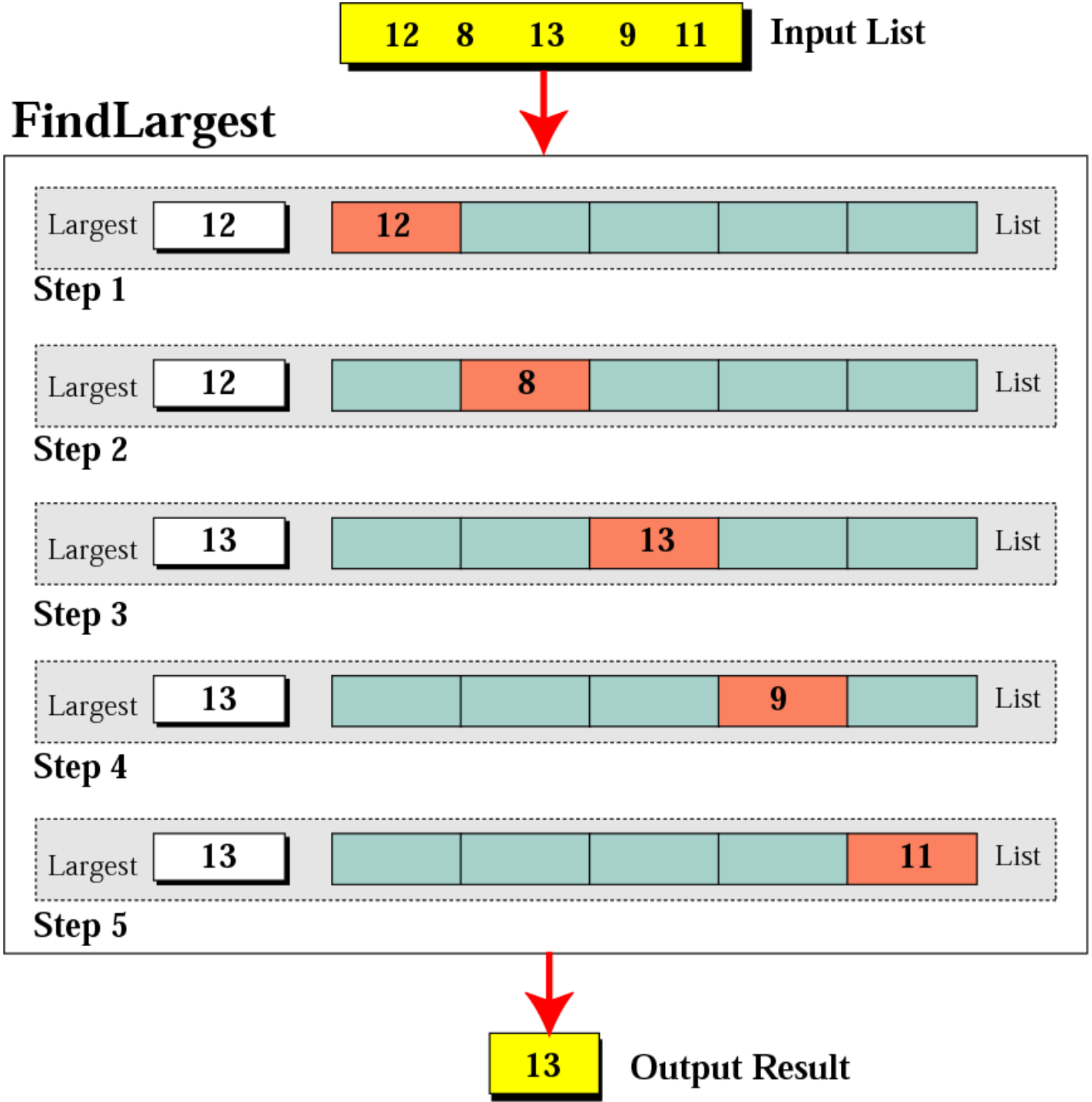
These properties are:

- ***Input:*** An algorithm has input values from a specified set.
- ***Output:*** From each set of input values an algorithm produces output values from a specified set.
- ***Definiteness:*** The steps of an algorithm must be defined precisely.
- ***Correctness:*** An algorithm should produce the correct output values for each set of input values.

Continue…..

- ***Finiteness.*** An algorithm should produce the desired output after a finite number of steps for any input in the set.
- ***Effectiveness***. It must be possible to perform each step of an algorithm exactly and in a finite amount of time.
- ***Generality.*** The procedure should be applicable for all problems of the desired form, not just for a particular set of input values

FindLargest



12 8 13 9 11 **Input List**

FindLargest

Set Largest to the first number.

Step 1

If the second number is greater than Largest, set Largest to the second number.

Step 2

If the third number is greater than Largest, set Largest to the third number.

Step 3

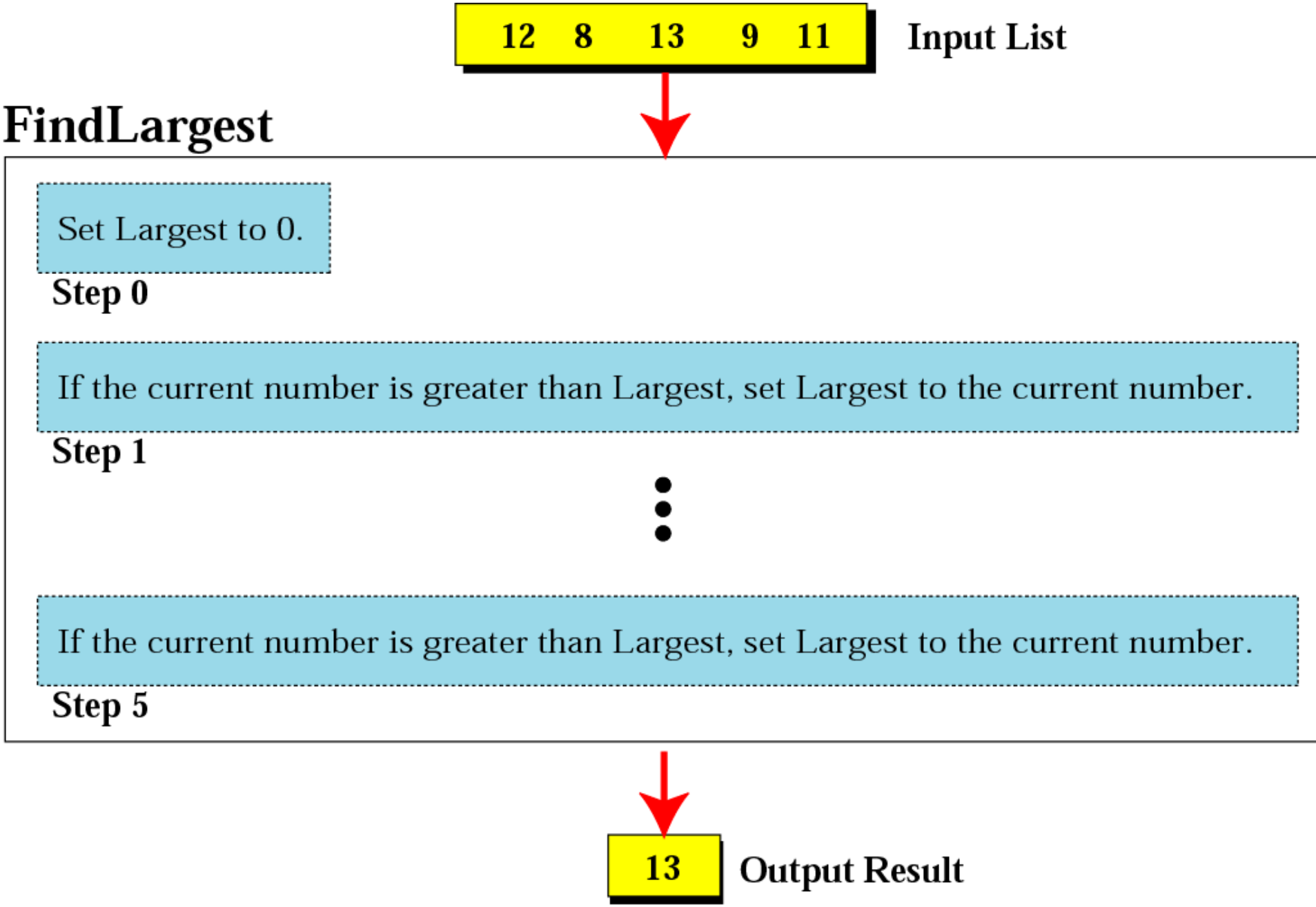
If the fourth number is greater than Largest, set Largest to the fourth number.

Step 4

If the fifth number is greater than Largest, set Largest to the fifth number.

Step 5

13 **Output Result**


12 8 13 9 11
Input List
FindLargest
Set Largest to 0.
Step 0
If the current number is greater than Largest, set Largest to the current number.
Step 1
If the current number is greater than Largest, set Largest to the current number.
Step 5
13
Output Result

Generalization

Input List

FindLargest

Set Largest to 0.

Repeat the following step *N* times:

If the current number is greater than Largest, set Largest to the current number.

Largest

Constructs of an Algorithm

- Computer scientists have defined three constructs for a structured program or algorithm.
- The idea is that a program must be made up of a combination of only these three constructs: sequence, decision (selection), and repetition.

Sequence

- An algorithm, and eventually a program, is a sequence of instructions,

do action 1
do action 2
……
……
do action n

a. Sequence

Decision

- Some problems can not be solved with only a sequence of instructions.
- Sometimes you need to test a condition.
- If the result of testing is true, you follow a sequence of instructions; if it is false, you follow a different sequence of instructions.
- This is called the decision (selection) construct.

Decision

if a condition is true,

then

do a series of actions

else

do another series of actions

b. Decision

Repetition

- In some problems, the same sequence of instructions must be repeated.
- You handle this with the repetition (loop) construct.

Repetition

while a condition is true,

do action 1
do action 2
. . .
. . .
do action n

c. Repetition

Programming Tools

- There are many tools which are used to convert algorithms into computer programs but the major one is:
- **Pseudocode** - Uses English-like phrases with some programming language terms to outline the program.

Pseudocode

- It is an English-like representation of the code required for an algorithm. It is part English and part structured code.
  - English part provides a relaxed syntax that is easy to read.
  - Code part consists of an extended version of the basic algorithmic constructs (sequence, decision and repetition).
- Until now, there is no standard for pseudocode.

Pseudocode example

Algorithm: Finding Smallest

Purpose: This algorithm finds the smallest number among a list of numbers.

Pre: List of numbers

Return: The Smallest

1. Set smallest to the first number

2. loop (up to end of list)

2.1 if (next number < smallest)

2.1.1 set smallest to next number

2.2 end

3. end loop

4. Return smallest

End finding Smallest

Pseudocode

```
action 1
action 2
   .
   .
   .
action n
```

a. Sequence

```
if (condition)
   then
      action
      action
      . . .
   else
      action
      action
      . . .
End if
```

b. Decision

```
while (condition)
      action
      action
      . . .
End while
```

c. Repetition